

INSTRUCTIONS

HOW THIS WORKBOOK WORKS

How to Use This Workbook

This is a working document, not a reading book.

Fill all sections during the workshop.
Avoid generic answers. Be specific.
Use real observations, not assumptions.

All final outputs will come from this workbook.

Important Rules:

- Minimum 5 real user interactions required
- Document actual observations and quotes
- Show what failed — not just what worked

Think clearly. Write concisely.

AI USAGE REMINDER

Use AI for -

- Structuring insights
- Expanding ideas
- Organizing notes

Do NOT use AI for -

- Fake research
- Fabricated validation
- Copy-paste solutions

01. PROBLEM & USER DEFINITION

Target User (Be Specific)

- Inpatients requiring continuous Intravenous (IV) therapy
- The clinical nursing staff managing their care.

Context (Where does this problem occur?)

- The problem actually occurs in resource-constrained inpatient clinical care environments(Ex:24/7 hospitals) and hospital floors with high patient-to-nurse ratios.

Why does this problem matter?

- Relying on manual observation for IV monitoring risks severe adverse events, such as air embolisms and undetected venous infiltration, which directly compromise patient safety.

Initial Assumptions (List at least 5)

- Manual IV monitoring is unreliable in high patient-to-nurse ratio environments.
- Current automated systems are purely reactive, only detecting failures after they happen.
- Flow anomalies can be mathematically predicted using real-time data.
- Edge AI can be processed locally on low-cost microcontrollers.
- Predicting the exact time-to-empty will prevent air embolisms.

Stakeholders (Who is involved?)

- Inpatients
- Nursing Staff
- Hospital Administration

Narrow the Problem (One clear line)

- Current IV monitoring relies on reactive systems or manual observation, failing to predict flow anomalies and risking severe adverse patient events like air embolisms.

02. RESEARCH & INSIGHT CAPTURE

Interview Log (Minimum 5 Users)

User	Insights
Nurse	Inability to predict empty times.
Head Nurse	Severe understaffing
Patient	Physical risk and fear
Doctor	Need for predictive data
Tech Lead	Hardware limitations

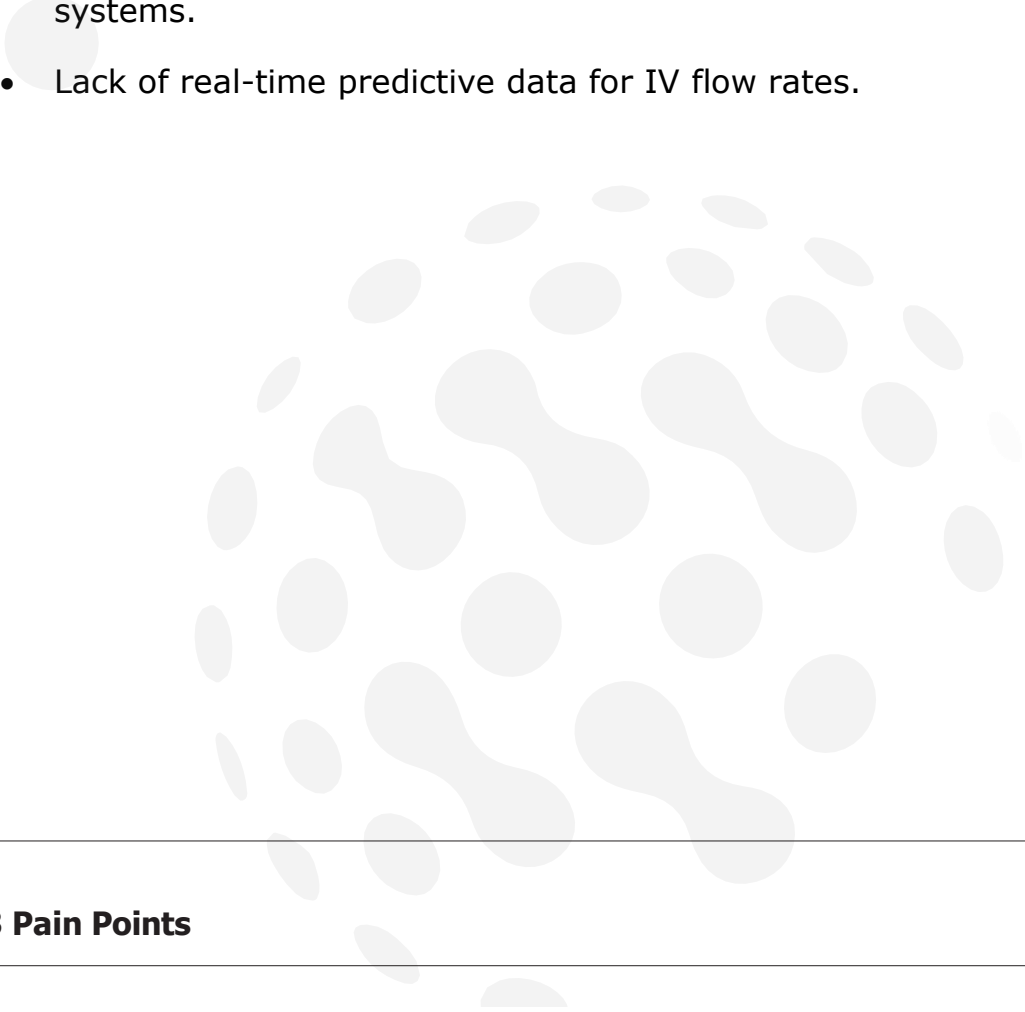
Direct Quotes

- **Nurse:** "By the time I notice the IV, it is already empty."
- **Head Nurse:** "We have too many patients to check every IV drip manually"
- **Patient:** "My relative's blood backed up into the tube before anyone noticed."
- **Doctor:** "We need a system that calculates flow rates in real-time, not just a threshold trigger."
- **Tech Lead:** "Centralized systems are too slow; we need processing at the bedside."

Observations

- Staff relies on reactive checking rather than proactive planning.
- High patient-to-nurse ratio makes manual monitoring impossible.
- Vulnerable to venous infiltration and air embolisms.
- Current tech is lacking time-series analysis.
- Highlights the need for decentralized Edge AI.

Pattern Clusters (Group Similar Findings)

- Extreme risk of air embolisms and venous infiltration.
 - Total reliance on manual observation or reactive threshold-based systems.
 - Lack of real-time predictive data for IV flow rates.
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Top 3 Pain Points

- High risk of severe adverse events (air embolisms) before staff can react.
- High patient-to-nurse ratios making manual IV monitoring dangerous.
- Lack of real-time time-series analysis to predict exact time-to-empty.

Insight Statement

User struggles with - preventing IV therapy failures (like air embolisms).

Not because - they are neglectful.

But because - they rely on manual observation and reactive systems instead of predictive, real-time flow analysis.

03. PROBLEM FRAMING

POV (Point of View)

User: An inpatient receiving continuous IV therapy and their assigned nurse.

Needs: A way to accurately predict flow anomalies and exact time-to-empty.

Because: Reactive systems and manual checks lead to a high risk of air embolisms and venous infiltration in resource-constrained environments.

HMW (How Might We)

- How might we use time-series analysis to predict exact IV empty times?
- How might we implement Edge AI directly at the patient's bedside?
- How might we eliminate air embolism risks before failures occur?

Final HMW (Select ONE)

How might we create a decentralized, predictive monitoring architecture using Edge AI to calculate exact time-to-empty and prevent severe adverse events like air embolisms?

04. IDEATION & SELECTION

Idea Generation (Crazy 8)

ESP32 microcontroller with a Load Cell for real-time weight tracking.	Miniaturized in-line turbine sensor that measures physical fluid flow.	Computer vision camera mounted on the pole to visually count IV drops.	Ultrasonic flow sensor clamped directly onto the IV tubing.
Wearable biometric patch to detect patient distress from venous infiltration.	Smart IV bags with built-in electronic shrinking sensors.	Robotic ward assistant that physically drives from bed to bed to check IVs.	Laser-based optical scanner measuring fluid density in the drip chamber.

Mission Impossible

- **The Idea:** A self-replenishing, wireless IV fluid teleportation system that directly hydrates the patient's bloodstream without bags, tubes, or needles, entirely eliminating the need for flow monitoring.
- **The Takeaway:** While impossible, it highlights that the ultimate goal is zero physical monitoring burden on the nursing staff and zero risk of physical complications (like air embolisms) for the patient.

Top 3 Ideas

- **Edge AI & Load Cell:** ESP32 microcontroller with a Load Cell to perform real-time time-series analysis on IV fluid weight.
- **Optical Drip Sensors:** Using infrared sensors clamped to the drip chamber to count individual drops.
- **Ultrasonic Tubing Sensor:** Using sound waves to measure fluid velocity directly inside the tube.

Impact vs Feasibility Grid

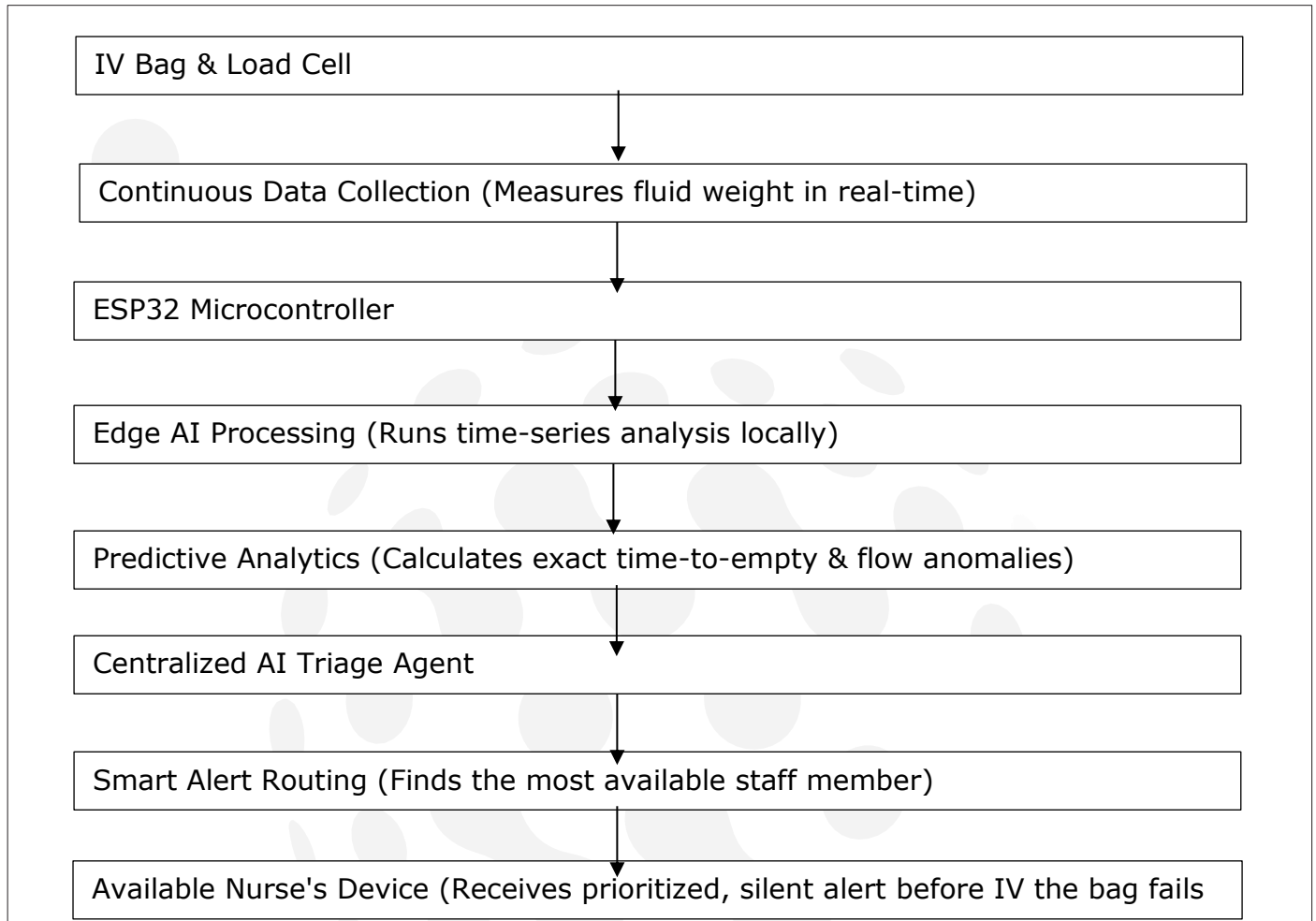
- Edge AI Load Cell: High Impact / High Feasibility (Highly accurate predictive modeling using low-cost hardware).
- Optical Drip Sensors: High Impact / Medium Feasibility (Susceptible to ambient light interference and requires line-of-sight).
- Ultrasonic Sensor: High Impact / Low Feasibility (Components are too expensive for resource-constrained environments).

Final Selected Concept

AI Driven IV Fluid Monitoring: A decentralized system using an ESP32 microcontroller and a Load Cell to run Edge AI, performing time-series analysis to predict exact time-to-empty and flow anomalies.

05. PROTOTYPE & TESTING

Sketch / Flow



User Interaction Steps/ Journey Map

- A Load Cell is attached to the IV bag to continuously measure fluid weight.
- The paired ESP32 microcontroller captures this continuous data stream locally.
- Edge AI performs real-time time-series analysis directly on the ESP32.
- The system successfully calculates a flow anomaly and the exact time-to-empty.
- An AI Triage Agent ingests this predictive data to proactively and silently alert the nurse before failure occurs.

Hypothesis

If we - implement Edge AI directly on an ESP32 microcontroller paired with a Load Cell

then - we can perform real-time time-series analysis to predict exact time-to-empty and eliminate air embolism risks

because - local, decentralized processing allows for proactive intervention rather than waiting for a reactive threshold trigger to fail.

Testing Feedback

Initial observations of traditional automated systems showed that broadcasting loud, reactive alarms to the entire hospital floor creates severe "alarm fatigue" among nursing staff. Furthermore, waiting for a failure to happen before alerting the staff puts patients at high risk for severe adverse events, such as air embolisms.

What Failed

Traditional threshold-based systems failed because they are strictly reactive, broadcasting non-differentiated alarms only *after* a critical failure has already occurred. This approach overwhelms resource-constrained floors with noise and fails to proactively protect the patient.

What Changed

We shifted from a reactive alarm model to a predictive architecture. By implementing Edge AI on an ESP32 with a Load Cell, the system now calculates exact time-to-empty before failures happen. We also integrated a centralized AI Triage Agent to dynamically route silent, prioritized alerts to the most available nurse.

06. FINAL SOLUTION SNAPSHOT

Final Concept (Max 5–6 lines)

A decentralized, predictive monitoring system for continuous IV therapy that implements Edge AI directly on an ESP32 microcontroller paired with a Load Cell. It conducts real-time time-series analysis to predict flow anomalies and exact time-to-empty before any failures occur. A centralized AI Triage Agent then ingests this predictive data. Finally, the agent dynamically routes prioritized, silent alerts directly to the most available staff member.

What Problem Does It Solve?

It actively eliminates the high risk of severe adverse events, such as air embolisms and undetected venous infiltration, which are caused by reliance on manual observation. Concurrently, it addresses dangerous "alarm fatigue" by replacing non-differentiated broadcast alarms with targeted, silent alerts, aiming for up to a 40% reduction in hospital floor noise.

Feasibility

- Operational - High. It integrates into resource-constrained clinical environments by optimizing nursing workflows and routing alerts only to the most available staff members.
- Financial - High. It utilizes highly accessible and cost-effective IoT Medical Devices, specifically ESP32 microcontrollers and standard Load Cells.
- Behavioral- High. It significantly improves staff well-being by reducing the severe stress and cognitive load associated with broadcast alarm fatigue.

Key Barrier

Seamlessly integrating the decentralized Edge AI nodes with the centralized AI Triage Agent to ensure predictive data is communicated without latency in a resource-constrained clinical environment.

If this page is strong, your project is strong.

Problem Clarity (1–2 lines) Define your problem in the simplest way possible

- Relying on manual observation and reactive threshold sensors for IV monitoring leaves inpatients highly vulnerable to severe adverse events like air embolisms.

Target User (Be Specific) Who exactly are you solving for?

- Inpatient clinical care patients requiring continuous IV therapy, and the nursing staff managing high patient-to-nurse ratios at facilities like Emergency Hospitals.

Core Insight

- Proactive patient safety requires predicting failures (exact time-to-empty) before they occur and routing silent alerts to specific staff, rather than waiting for a failure to trigger a reactive, non-differentiated alarm.

What did you learn from users that changed your thinking?

- We learned that waiting for an IV bag to empty before taking action is fundamentally unsafe. To truly safeguard patients, the monitoring system must shift from being reactive to being predictive by calculating flow anomalies before a critical failure occurs.

Key Pain Points (Top 3)

- High risk of severe adverse events, such as air embolisms and undetected venous infiltration.
- Complete inability of current systems to predict exact time-to-empty in advance.
- High patient-to-nurse ratios making manual IV monitoring impossible and dangerous.

Your “Aha Moment” When did things click for you?

- Realizing that by running Edge AI directly on a low-cost ESP32 microcontroller paired with a Load Cell, we can perform real-time time-series analysis to predict exact empty times before failures happen.

Final HMW (How Might We)

- How might we utilize Edge AI and real-time time-series analysis to predict exact IV time-to-empty, thereby eliminating air embolism risks and safeguarding patients?

Your final problem framing

- How might we utilize Edge AI and real-time time-series analysis to predict exact IV time-to-empty, thereby eliminating air embolism risks and safeguarding patients?

Solution Summary (Max 3 lines) What are you building?

- A decentralized IV monitoring architecture utilizing an ESP32 and Load Cell to perform predictive Edge AI time-series analysis. It calculates the exact time-to-empty and uses a centralized AI Triage Agent to dynamically route alerts to available staff before a failure occurs.

Why THIS Solution? (Decision Logic)

- It directly solves the root cause of IV therapy risks (lack of proactive data) by replacing delayed, reactive threshold sensors with a highly accurate, decentralized predictive architecture.

What's unique or better?

- Instead of just triggering when a bag is completely empty, it moves the AI processing to the "Edge" (the patient's bedside) to perform continuous time-series analysis and predict failures in advance.

What proof do you have that this works?

- By performing time-series analysis on real-time Load Cell data, the architecture can successfully calculate flow anomalies and exact time-to-empty to proactively eliminate air embolism risks.

Biggest Assumption You Took

- Assuming the Load Cell data will remain stable and the Edge AI models will stay accurate despite patient movement artifacts or vibrations in the

What Failed? (Be Honest) What didn't work?

- Traditional automated systems failed to protect patients because they rely on reactive thresholds, only detecting a problem *after* a critical failure has already happened.

Feasibility Check (Can this be built in real life?)

✓ Yes ☐ Maybe ☐ No ...Why? : The system leverages highly accessible, cost-effective IoT medical device hardware combined with proven Edge AI time-series data modeling.

Impact (If this works...) What will change for users?

- It will completely eliminate the risk of air embolisms, transforming IV monitoring from a risky manual burden into a highly accurate, automated, predictive science.

One Line Pitch (Explain your idea in one powerful sentence)

A decentralized Edge AI system that uses real-time time-series analysis to predict exact IV empty times, eliminating air embolism risks and proactively safeguarding patients.

[illegible]